

Enterprise IT & Cloud Infrastructure Manual

Standard Operating Procedures, Troubleshooting & Deployment Protocols

1. API Gateway & Authentication Protocols

All internal microservices interact via OAuth 2.0 bearer tokens issued by the Central Authorization Server. Rate limits are strictly enforced at 1,000 requests per minute per IP address for standard tiers, and 10,000 requests per minute for enterprise service accounts.

```
Authorization: Bearer <YOUR_JWT_TOKEN>
X-RateLimit-Limit: 1000
X-RateLimit-Remaining: 998
```

2. Production Incident Escalation Matrix

In the event of a Severity 1 (P1) outage affecting core database clusters or vector search indexes:

- **0-15 Minutes:** On-call DevOps Lead acknowledged alert and creates Incident Channel #inc-p1.
- **15-30 Minutes:** Notify Site Reliability Engineering (SRE) Manager and deploy fallback database read-replicas.
- **30+ Minutes:** Initiate cross-region failover via AWS Route 53 DNS routing.

3. Database Backup & Vector Storage Retention

Vector database indexes hosted on Pinecone and ChromaDB undergo daily automated snapshots at 02:00 UTC. Snapshot retentions are maintained for 30 consecutive days in encrypted S3 cold storage buckets.